

COMMAND NEXUS · CIT 140 · WEEK 1

# Problem Brief

Meridian Forge Robotics · The Process Below

|                                     |                                  |                                   |
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| <b>LEARNER</b><br>Stephania Malagon | <b>ARTIFACT</b><br>Problem Brief | <b>STATUS</b><br>Working document |
|-------------------------------------|----------------------------------|-----------------------------------|

SECTION 01

## Visible Failure

MR-17 began repeating Test K-44, which was retired 186 days before the incident. K-44 was removed from the active workflow but records do not confirm that every copy, service, or schedule was removed. The operator command to stop MR-17 was validly formatted and returned, "ACTIVE\_INSTRUCTIONS\_PENDING".

SECTION 02

## Verified Facts

1. Mara Ilyan (Production Operations Manager) - Said that Nia reported that MR-17 was behaving normally before the incident.
2. Maintenance Record: The record does not confirm that every copy, service, or schedule was removed.
3. Elias Chen (Robotics Systems Engineer) - The source label is legacy\_sync.exe .

**SECTION 03**

## Assumptions and Open Questions

1. Assumption: The maintenance records do not confirm that K-44 has been removed from every possible area MK-17 is in.
2. Open Question: A controlled restart may clear the active state.

**SECTION 04****System Environment Map****SECTION 04A****People and Roles**

Nia Voss - Night QA Supervisor  
Mara Ilyan - Production  
Operations Manager  
Elias Chen - Robotics Systems  
Engineer

**SECTION 04B****Processes and Rules**

Production Operations  
Manager controls what work  
runs in the QA cell, the  
operating schedule, and  
software history.

**SECTION 04C****Information and Data**

QA lab, Records Terminal, Test  
Catalog, Operation Schedule,  
Software History

**SECTION 04D****Technology and Tools**

MR-17, QA Operations  
Interface, The Robot

**SECTION 04E****Environment and Constraints**

The only limit is time before the  
Robot restarts and clears any  
information needed.

**SECTION 05**

## Investigation Boundary

**Inside the Investigation**

The team needs to investigate the sequence of commands that led to MR-17 repeating retired Test K-44.

**Outside for Now**

MR-17 repeating retired Test K-44 is a fact and does not need additional investigation.

**SECTION 06**

## Organizational Impact

The QA cell has been isolated so only the current team is affected.

**SECTION 07**

## Problem Statement

There is an informational gap of the software history and maintenance record. The team does not have any info on the software history and maintenance record does not verify that Test K-44 was removed from every aspect and area of the robot.

**SECTION 08**

## Next Investigation Step

The team needs to determine what legacy\_sync.exe is and its affect on the robot. That specific unknown file might be the cause of MR-17 breaking out of the scheduled sequence of commands and repeating Test K-44 instead.

**SHARED ARTIFACT**

## Working Document Link

[Not entered](#)

**WEEK 1 STOPPING RULE**

## Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.